

CURRICULUM VITAE

Aaron Kuang

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Citizenship: USA

EDUCATION

University of California, Irvine (UCI)

Mar 2026

B.S. in Computer Science

GPA: 4.00/4.00

Campuswide Honors Collegium, ICS Honors Program.

Honors Thesis: Help-seeking Behaviors in Algorithms Courses

RESEARCH INTERESTS

Theoretical Computer Science (TCS) Education

Interested in discovering and alleviating barriers to success in TCS courses. Particularly curious about how psychological factors — such as sense of belonging, self-efficacy, and impostor syndrome — manifest while learning TCS and algorithms. Emphasis on incorporating student sentiments, boosting engagement, and broadening participation in TCS.

Human–AI Interaction

Interested in understanding how people — especially students, teachers, creatives, and researchers — interact with and perceive generative AI. Emphasis on issues related to ethics, cognitive offloading, and effects of integration. Investigating how generative AI has transformed learning environments and changes learning outcomes.

Algorithm Design

Intrigued by reductions and the application of subroutines as building blocks for larger algorithms. Curious about computing paradigms that can provide speedups over classical frameworks, such as parallel and quantum paradigms. Especially interested in fleshing out complexity classes and tackling NP-complete problems and problems in TFNP.

RESEARCH EXPERIENCE

Help-seeking Behaviors in Algorithms Courses

Sep 2025 – Present

Advisor: Prof. Michael Shindler

Co-PI/Mentor: Prof. Stephen Tsung-Han Sher

- Co-authored a qualitative research study that found that students enrolled in algorithms courses at one public R1 university weighed trade-offs between convenience, quality, and perceived effect on grades when seeking help.
- Coordinated and scheduled 22 interviews between members of the research group and study participants by collecting interviewer and interviewee availability.
- Oversaw the anonymization and cleaning of student transcripts, then assisted with qualitative coding, phenomenographic analysis, and paper writing.
- Attended two weekly meetings to discuss prior literature on help resource usage, lay down expectations for the next meeting, debrief on current progress, and identify cross-cutting trends seen in the transcripts.

Resisting Humanity’s Mechanization in *Fahrenheit 451* Apr 2023 – Jun 2023
Mentor: Dr. Katherine Raymond Walsh

- Wrote an independent literary analysis research paper examining how the technological dystopia of *Fahrenheit 451* affects its citizens’ daily lives.
- Discussed how anti-intellectualism and over-reliance on technology has eroded the characters’ need to think critically, rendering them blind to the horrors of the world.
- Reflected on how modern feats of technology—social media, virtual reality, and artificial intelligence—can erase our individuality and humanity if left unchecked.

TEACHING EXPERIENCE

Learning Assistant (University of California, Irvine):

Design and Analysis of Algorithms

Spring 2025, Fall 2025

Data Structure Implementation and Analysis

Fall 2024

- Facilitated active learning in lectures and discussions by supporting students during in-class activities designed to reinforce concepts in data structures and algorithms.
- Supported exam review sessions meant to help students with recursion, graphs, and proofs. Presented slideshows and provided study tips.
- Attended weekly meetings with course staff to discuss pedagogical strategies, potential weak points in students’ understanding, and course logistics.
- Helped administer exams in the algorithms course — proctored and scanned exams.

AWARDS, HONORS, AND SCHOLARSHIPS

Outstanding Contribution to Research by an ICS Undergraduate Jun 2026

- Nominated by a UCI faculty member for contributions to research.

UCI PBK Fall Book Award

Nov 2023

- Awarded to UCI sophomores who demonstrate academic excellence.

Victoria Alegria Tracy Endowed Scholarship

Sep 2023

- \$3,000 awarded to four UCI undergraduates annually.

UROP Humanities Core Research Paper Award Nomination

Jun 2023

- Nominated by Dr. Katherine Walsh for a “superior research paper.”

AFFILIATIONS

Phi Beta Kappa

Since Jun 2026

- Nominated by UCI’s Mu chapter as part of the top 5% of graduating seniors.

UCI ICS Honors Program (ICSHP)

Sep 2025 – Mar 2026

UCI Campuswide Honors Collegium (CHC)

Sep 2022 – Mar 2026